

Question Paper

Exam Date & Time: 24-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Computer Vision [ECE_4410]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) With a neat schematic, explain the image formation process in human eye? (5)
A) Course Outcome: "CO1" / Cognitive Level: "Understand"
B) Demonstrate the process of sampling and quantization with respect to acquisition of digital images? (5)
Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Demonstrate the concept of order statistics filters used in digital image processing. Explain the computational steps used in median filtering. (5)
A) Course Outcome: "CO2" / Cognitive Level: "Application"
B) Explain the sharpening filter used in processing digital images with an example. State any one application problems where sharpening filters can be used. (5)
Course Outcome: "CO2" / Cognitive Level: "Understand"
- 3) Illustrate the edge detection problem in digital images. Why derivative filters are used in edge detection. (5)
A) Course Outcome: "CO3" / Cognitive Level: "Application"
B) Using basic principles, derive the mask coefficients of a Prewitt edge detector. (5)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Deduce the steps involved in forward selection method used in feature selection using fundamental principles? (5)
A) Course Outcome: "CO4" / Cognitive Level: "Analysis"
B) Demonstrate the Bayes classifier used for pattern classification with the necessary computational steps, assuming binary pattern classification problem. (5)
Course Outcome: "CO4" / Cognitive Level: "Application"
- 5) Differentiate between multilayer perceptron and support vector machine used in classification? (5)

A) Course Outcome: "CO5" / Cognitive Level: "Analysis"

B) Explain the interpretation of using kernel functions in learning SVM classifier. Illustrate with one example kernel function, with its mathematical formulation? (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

ECE_4410:Computer Vision

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

ECE_4410:Computer Vision

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	10(20%)
3	Application	5	25(50%)
4	Analysis	3	15(30%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)